

SOC Network Traffic Investigation Labs - Project Report (Labs 01-12)

This report documents twelve hands-on SOC Analyst home labs using Wireshark.

Lab01: Environment Setup – Built Ubuntu and Kali Virtual Machines, configured networking and installed Wireshark.

Lab02: ARP & ICMP – Investigated MAC resolution and ping traffic.

Lab03: DNS Investigation – Analyzed DNS queries and responses.

Lab04: TCP Three-Way Handshake – Observed SYN, SYN-ACK and ACK packets.

Lab05: HTTP Traffic Analysis – Investigated HTTP requests, responses and status codes.

Lab06: HTTPS/TLS Analysis – Examined TLS handshake and encrypted communication.

Lab07: FTP Traffic Analysis – Studied FTP authentication and file transfers.

Lab08: Nmap TCP SYN Scan – Captured reconnaissance traffic and SYN scanning behavior.

Lab09: SSH Traffic Analysis – Examined SSH version exchange, key exchange and encrypted packets.

Lab10: DHCP Analysis – Captured DHCP Discover, Offer, Request and ACK.

Lab11: SMB Traffic Analysis – Configured Samba, generated SMB2 file operations and analyzed file-sharing traffic.

Lab12: Telnet Traffic Analysis – Observed insecure plaintext remote access and discussed security risks.

Learning Outcomes:

- Packet capture and protocol analysis.
- Wireshark display filters.
- Identifying reconnaissance activity.
- Understanding authentication and encryption.
- Practical SOC investigation skills.

Conclusion:

The project provides practical experience with protocols that SOC analysts monitor daily, helping develop skills in traffic analysis, incident investigation, and threat detection.